

TABLE II
INDEPENDENT INTERNAL-TEST PERFORMANCE OF THE SIX IMPROVED DEEP LEARNING MODELS

Rank	Model	Accuracy (%)	Macro F1 (%)	Balanced Acc. (%)	MCC	Log Loss
1	ConvNeXt-Tiny	99.9583	99.9490	99.9333	0.999526	0.004070
2	ResNet50	99.8749	99.8555	99.8912	0.998579	0.005143
3	InceptionNet	99.7916	99.7964	99.8148	0.997631	0.011315
4	EfficientNet-B0	99.7916	99.7004	99.7747	0.997630	0.010794
5	MobileNet	99.6248	99.5705	99.5807	0.995737	0.012934
6	VGG16	99.5832	99.4848	99.3507	0.995262	0.027355

Note: Results are from the untouched PlantVillage independent test set ($n = 2,399$). Only the Improved training configuration is shown. Macro F1 was the primary ranking metric; bold indicates the selected final model.